A Shapley-value framework for quantifying module contributions in context-augmented decision support: application to parcel-level land management

Modular architecture

Satellite



Vegetation indices

Orthophoto



High-res imagery

RAG



Knowledge base

Dataset construction

(1) Parcel selection
(2) Land Use-related query elaboration
(3) Response generation

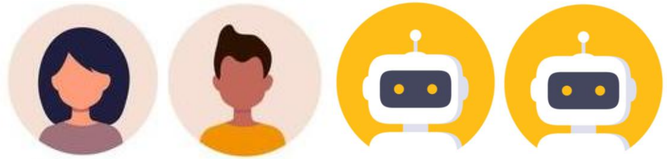
143 samples

Methodology stages

Broad

Ablation

Panel of judges



Hybrid Human-in-the-Loop

Metrics

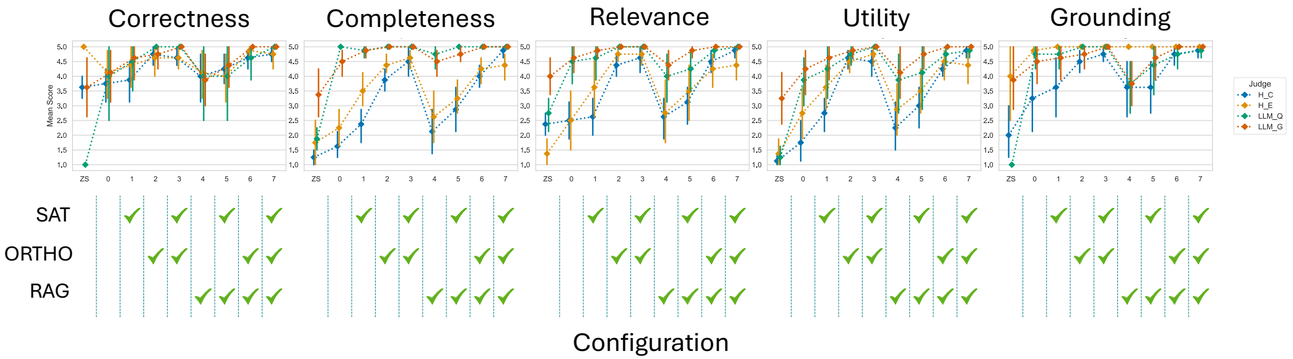
- ✓ Correctness
- ✓ Completeness
- ✓ Relevance
- ✓ Utility
- ✓ Grounding

1-5 Likert scale

Analysis & key findings

Collaborative game theory framework for marginal contributions

SHAPLEY Values



- Validated architecture
- Visual context crucial
- Marginal contributions successfully quantified
- Reproducibility
  Dataset is available